Region shared(stateful, server-authoritative)

Simulation service (authoritative world/ routes)

player command ingest(from WS to Flight request)

Enqueue Flight requests

RequestQueue (BlockingQueue<FlightRequest>)

~~RequestBus~~

Blocking take() Dispatcher

Distpatcher

sc schedul servicing ~~tasks~~

PlaneServisingpool (Threadpool) ServicingManager+ StandardPlaneServicing

Dispatch backlog loggin

Tick Loop (Ticker)

World Init (WorldBuilder)

build airports/plains ~~worldBuilder~~

on complete: mark serviced or available

Gameplay event(analytics)

Map/state(Airports, Planes) (in-memory, hot cache)

compute details from SIM_STATE

snapshot/Deltas Publisher (to client via WS)

player commands (create route, dispatch, chat)

Maps to my code:
-SimulationController or chestrate shared service
-WorldBuilder builds airports/plans and GridAreaIcons
-RequestBus is the blocking queue used between producer/dispatcher
-Dispatcher consumes requests and calls "dispatchOrBacklog"
- Servisingmanager wraps a threadpool for StandardPlaneservicing

Client

Destop client

Mobile client

websockets(state,cmds)

HTTPS(login,purcheses, profile)

HTTPS

assets,patches

websockets(state,cmds)

assets,patches

Regional Edge

Realtime Gateway (WebSockets, push)

API gateway/Auth (HTTPS, JWT/OIDC)

CDN(Assets/Pathches)

Low latency
- regional presence
- Websocket push
- thin details

push snapshot details to client

auth, profiles, friends/history

purchase APIs, inventory ops

pick regions shard

risk check

feature flags/config

Low latency
- regional presence
- Websocket push
- thin details

Regional services

Identity and profile

chat and social

Economy and inventary (Transaction, Idempotent API)

Matchmaking/Region Directory

Anti-cheat/ Risk

Content/config (Feature Flags/ remote config)

accounts/profiles

rich, cheat events

Regional services

Redis/ KV Cache (Hot sim state/ sessions)

Telemetry/ Logs/Tracers (observerbility)

Rational DB (PostgreSQL/MySQL) (Accounts, Economy, Recipts)

EventBus(Kafka/ SNS+ SQS)

ACID money items

backups/snapshots

Object storage (Assets, Backups)

Economy integrity:
-ACID transactions
-idempotent APIs
- durable audit logs

content bundles